

FINAL PROJECT REPORT GUIDELINE

Your final report should demonstrate that your team can use financial data, analytical methods, and Python to investigate a meaningful financial problem and produce a reliable, interpretable, and decision-relevant conclusion. The report is not a record of everything you tried.

THE REPORT MUST ANSWER

What financial problem did you investigate? What does the evidence show? How reliable is the analysis? What are the financial implications?

RECOMMENDED PROJECT STORY

Financial Problem

Data

Preparation

EDA

Feature Engineering

Method / Model

Validation

Financial Interpretation

Recommendation

REPORT STRUCTURE – TEN SECTIONS

01 Executive Summary

02 Problem & Objectives

03 Data Understanding

04 Data Preparation

05 EDA & Financial Analysis

06 Feature Engineering & Method

07 Results & Evaluation

08 Financial Interpretation

09 Recommendation / Decision Implication

10 Limitations & Conclusion

LENGTH

Approximately 15–20 pages,
excluding appendix and references.

DELIVERABLES

Report PDF, notebooks / source
code, README, dataset or
instructions, final presentation,
optional dashboard.

NON-NEGOTIABLES

No look-ahead bias, no data leakage,
no shuffled time series, always
a benchmark.

THE FINAL TEST

Correct Data + Correct Method + Correct Validation + Financial Interpretation = Good Financial Data Analytics

1 GOAL

Your final report should demonstrate that your team can use **financial data, analytical methods, and Python** to investigate a meaningful financial problem and produce a **reliable, interpretable, and decision-relevant conclusion**.

The report is **not a record of everything you tried**. It should clearly answer:

What financial problem did you investigate? What does the evidence show? How reliable is the analysis? What are the financial implications?

RECOMMENDED PROJECT STORY

Financial Problem → Data → Preparation → EDA → Feature Engineering → Method/Model → Validation → Financial Interpretation → Recommendation

2 FINAL REPORT STRUCTURE

SECTION	WHAT TO INCLUDE
1. Executive Summary	Financial problem, data, approach, key findings, model/result, financial implication, recommendation
2. Problem & Objectives	Financial/business context, research question, objectives, success criteria
3. Data Understanding	Data sources, period, frequency, variables, target, financial meaning, data quality issues
4. Data Preparation	Missing values, duplicates, dates, outliers, corporate actions, merging, frequency alignment, transformations
5. EDA & Financial Analysis	Trends, returns, volatility, distributions, correlations, regimes, anomalies, financially meaningful patterns
6. Feature Engineering & Method	Financial features, baseline, selected models/methods, assumptions, rationale, validation strategy
7. Results & Evaluation	Appropriate metrics, benchmark/model comparison, train vs. test results, errors, robustness
8. Financial Interpretation	What results mean financially, important drivers, risk/return implications, error analysis
9. Recommendation / Decision Implication	Evidence → Interpretation → Financial implication → Recommended action
10. Limitations & Conclusion	Assumptions, data/model limitations, risks, what was learned, possible improvements

Recommended length: approximately **15–20 pages**, excluding appendix and references.

3 WHAT MAKES A STRONG FDA PROJECT?

3.1 FINANCIAL PROBLEM THINKING

Can you formulate a meaningful financial question? Examples:

- Can we predict customer default or churn?
- Can historical information help forecast volatility?
- Can alternative data improve return prediction?

- Can portfolio risk be reduced without sacrificing expected return?
- Can transaction patterns identify potential fraud?

The question should be **answerable with the available data**.

3.2 DATA THINKING

Do you understand what each variable means financially? Pay particular attention to:

time period and data frequency	missing observations	outliers
inconsistent dates	corporate actions	market calendars
look-ahead bias	survivorship bias	data leakage

Good financial models start with good financial data.

3.3 ANALYTICAL THINKING

Choose methods because they fit the problem—not because they are complicated. A useful comparison is **Financial Benchmark → Simple Model → More Advanced Model**. Examples:

- Historical mean → Linear Regression → Random Forest
- Buy-and-hold → Portfolio strategy
- Naive volatility forecast → Statistical / ML model
- Majority-class prediction → Logistic Regression → Tree-based model

3.4 CRITICAL THINKING

Do not ask only: “Is the model accurate?” Also ask:

Does it generalize? Is the result financially meaningful? Would it still be useful outside the sample?

Test assumptions, investigate errors, and challenge your own conclusions.

3.5 FINANCIAL COMMUNICATION

A reader should understand: **What happened → Why it matters → How confident we are → What decision follows**.

4 IMPORTANT EXPECTATIONS

4.1 EDA

Every important figure should communicate **Question → Evidence → Observation → Financial Interpretation**. For example:

QUESTION	Does higher volatility occur during market declines?
EVIDENCE	Plot return against rolling volatility.
OBSERVATION	Large negative returns are associated with higher volatility.
INTERPRETATION	Downside markets may exhibit asymmetric risk.

Do not include charts simply because they look interesting.

4.2 TIME SERIES & VALIDATION

Financial data are usually **time dependent**. Therefore:

Do not randomly shuffle time-series observations unless you can justify it.

Prefer **Past → Training → Validation → Future Test**, or use **walk-forward validation**. Be especially careful about **look-ahead bias**: information used to predict time $t+1$ must have been available at time t .

4.3 MODELING

Always establish a reasonable **baseline or benchmark**. Use evaluation metrics appropriate to the problem.

CLASSIFICATION

Precision, Recall, F1, ROC-AUC, PR-AUC. Consider the financial cost of False Positive $\neq$ False Negative. For example, failing to detect fraud may have a different cost from incorrectly flagging a legitimate transaction.

REGRESSION / FORECASTING

MAE, RMSE, R^2 , out-of-sample R^2 . But statistical accuracy alone is not enough. Ask: does a better forecast produce a better financial decision?

PORTFOLIO / INVESTMENT

Return, Volatility, Sharpe Ratio, Maximum Drawdown, Calmar Ratio. When annualizing quantities, state your assumptions and apply them consistently.

5 INTERPRETATION

Do not stop at "Random Forest achieved an RMSE of 0.035." Explain:

How much better is this than the benchmark? Where does the model fail? Which variables matter? Is the improvement financially significant?

Separate three ideas: **Prediction $\neq$ Explanation $\neq$ Causation**. A model may predict well without proving that a variable **causes** the financial outcome.

6 STATISTICAL VS. FINANCIAL SIGNIFICANCE

A strong FDA project should distinguish **statistical performance** — does the method produce reliable predictive evidence? — from **financial / economic significance** — is the improvement large enough to matter financially?

For an investment strategy, consider whether results remain attractive after realistic assumptions such as transaction costs, turnover, liquidity, slippage and risk exposure. A small prediction improvement may not translate into useful financial value.

7 REQUIRED DELIVERABLES

1. **Final Report – PDF**
2. **Python notebooks / source code**
3. **README** explaining how to reproduce the analysis
4. **Dataset or clear instructions for obtaining it**
5. **Final Presentation (TBA)**
6. **Optional dashboard / application / financial decision tool**, if appropriate

Your analysis should be **reproducible**. Avoid undocumented manual data editing. Clearly state: data sources, data period, assumptions, random seeds where relevant, model parameters, and evaluation procedure.

8 FINAL CHECK BEFORE SUBMISSION

Ask your team:

Can a financial analyst or decision-maker who did not work on this project understand the problem, trust our evidence, and know what financial conclusion follows?

THE OBJECTIVE IS NOT

“Build the most advanced model.”

THE OBJECTIVE IS

“Use financial data correctly to produce reliable evidence, meaningful financial insight, and a defensible decision.”

A USEFUL FINAL TEST

Correct Data + Correct Method + Correct Validation + Financial Interpretation = Good Financial Data Analytics

A technically sophisticated model with data leakage, incorrect validation, or weak financial interpretation is not a strong project.